🕵️ SOC INVESTIGATION REPORT

Brute Force Login Detection

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1. Executive Summary

This report documents the investigation of suspicious authentication activity identified as a potential brute-force attack. The analysis was conducted using Splunk SIEM and Windows Event Logs to determine the nature, severity, and impact of the activity.

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2. Incident Description

A high volume of failed login attempts was observed targeting a specific user account. The pattern of repeated authentication failures suggested automated password guessing attempts.

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3. Investigation Methodology

The investigation followed standard SOC procedures:

- Reviewed authentication logs for failed login events (Event ID 4625)
- Identified source IP addresses associated with repeated attempts
- Correlated login activity over time
- Checked for successful login events (Event ID 4624)
- Analyzed behavior for signs of compromise

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4. Evidence & Findings

4.1 Failed Login Activity

- Multiple failed login attempts recorded within a short time frame
- Same source IP responsible for majority of attempts

4.2 Source Analysis

- Single IP address identified as primary source of login attempts
- Behavior consistent with brute-force attack pattern

4.3 Account Targeting

- Specific user account repeatedly targeted

4.4 Successful Login Check

- No successful login detected (update if applicable)

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5. Analysis

The repeated authentication failures, combined with consistent targeting of a single account from one source, strongly indicate a brute-force attack attempt. The absence of successful login suggests the attack was unsuccessful but still represents a security risk.

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6. MITRE ATT&CK Mapping

- **T1110 — Brute Force**

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7. Risk Assessment

- **Severity:** Medium (or High if login succeeded)
- **Impact:** Potential account compromise
- **Likelihood:** High (active attack behavior observed)

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8. Recommendations

- Block the malicious IP address
- Enforce strong password policies
- Enable Multi-Factor Authentication (MFA)
- Implement account lockout thresholds
- Continuously monitor authentication logs

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9. Conclusion

The investigation confirms that the observed activity is consistent with a brute-force attack. Although no successful compromise was identified, preventative measures are required to mitigate future risks.